

Week 3: Seller and Inventory Management System

Scenario:

The e-commerce company sells products supplied by multiple sellers. The system should maintain seller details, track products supplied by each seller, and monitor inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25),Email VARCHAR(30), Phone number(10), Address VARCHAR(45));
Table created.

SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7),
 2 Product_ID VARCHAR(7), Seller_ID VARCHAR(7),
 3 Stock_Quantity number, Stock_status VARCHAR(11),
 4 Last_Updated DATE);
Table created.
```

2. Establish Relationships

Seller and Product

One seller can supply multiple products.

One product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is one to many

Product and Inventory

Each product has inventory information.

Inventory tracks available stock.

Relationship:

Product → Inventory is one to one

Primary Key added for Both Tables:

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);
Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD
 2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);
Table altered.
```

```
SQL> ALTER TABLE Inventory ADD
 2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);
Table altered.
```

Constraints to make Unique Email ID:

```
SQL> ALTER TABLE Seller ADD
      2 CONSTRAINT Un_mail UNIQUE(Email);
```

To make Email NOT NULL

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;
Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory ADD
      2 CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);
Table altered.
```

Insert Values to Seller Table

```
SQL> INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
      2 ;
Enter value for sellid: S00001
Enter value for sellername: Vani Enterprises
Enter value for email: mdvanienter@gmail.com
Enter value for phone: 8765432109
Enter value for address: 1, Anna Nagar, Chennai
old 1: INSERT INTO Seller VALUES('&SellID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00001','Vani Enterprises','mdvanienter@gmail.com',8765432109,'1, Anna Nagar, Chennai')
1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S00001	Vani Enterprises	mdvanienter@gmail.com	8765432109
1, Anna Nagar, Chennai			
S00002	SD & CO	sellerD827@gmail.com	9876543012
62, Thirumal Nagar, Thanjavur			
S00003	AtoZ	atozmarket@gmail.com	7890654312
90, George Town, Chennai			
SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S00004	DSVV Manufaturers	dsvvmanuf142@gmail.com	7824003009
23, Thillai Nagar, Trichy			

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',
  2  &Stock,'&Stock_status',TO_DATE('&Last_upd','YYYY-MM-DD'));

Enter value for invid: Inv0001
Enter value for prodid: P100001
Enter value for sell: S00001
old   1: INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',
new   1: INSERT INTO Inventory VALUES('Inv0001','P100001','S00001',
Enter value for stock: 4
Enter value for stock_status: Available
Enter value for last_upd: 2026-09-12
old   2: &Stock,'&Stock_status',TO_DATE('&Last_upd','YYYY-MM-DD'))
new   2: 4,'Available',TO_DATE('2026-09-12','YYYY-MM-DD'))

1 row created.
```

```
SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
-----
Inv0001 P100001 S00001          4 Available  12-SEP-26
Inv0002 P100001 S00002          3 Available  13-AUG-26
Inv0003 P100002 S00003          4 Unavailable 15-SEP-26
Inv0004 P200002 S00004          3 Available  14-SEP-26
```

Report 1: Seller-wise Product Report

Display:

- Seller name
- Number of products supplied
- Available stock

```
SQL> SELECT s.Seller_Name,
  2  COUNT(i.Product_ID) AS Number_of_Products,
  3  SUM(i.Stock_Quantity) AS Available_Stock
  4  FROM Seller s
  5  JOIN Inventory i ON s.Seller_ID = i.Seller_ID
  6  WHERE i.Stock_Status = 'Available'
  7  GROUP BY s.Seller_Name;

SELLER_NAME                NUMBER_OF_PRODUCTS AVAILABLE_STOCK
-----
Vani Enterprises              1              4
SD & CO                      1              3
DSVV Manufaturers            1              3
```

Use Joins to get details from Seller and Inventory Table.

Report 2: Stock Availability Report

Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status
2 FROM Product p
3 JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATU
Laptops	4	Available
Laptops	3	Available
Head Phones	4	Unavailable
Kids Shoes	3	Available

Report 3: Inventory Analysis

Find:

- Total products available

```
SQL> SELECT COUNT(*) AS Total_Available_Products
2 FROM Inventory
3 WHERE Stock_Status = 'Available';
```

TOTAL_AVAILABLE_PRODUCTS
3

- Products out of stock

```
SQL> SELECT COUNT(*) AS OutOfStock_Products
2 FROM Inventory
3 WHERE Stock_Status = 'Out of Stock';
```

OUTOFSTOCK_PRODUCTS
0

- Highest stocked products

```
SQL> SELECT p.Product_Name, i.Stock_Quantity
2 FROM Product p
3 JOIN Inventory i ON p.Product_ID = i.Product_ID
4 WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

PRODUCT_NAME	STOCK_QUANTITY
Laptops	4
Head Phones	4

- Average inventory quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
2 FROM Inventory;
```

AVERAGE_INVENTORY
3.5